VIVEKANAND EDUCATION SOCIETY'S INSTITUTE OF TECHNOLOGY (VESIT)

Department of Automation and Robotics / Instrumentation | University of Mumbai

BATTERY STATE ESTIMATOR: AN ESN-BASED ALTERNATIVE TO EXTENDED KALMAN FILTER AND COULOMB COUNTING

Final Year Bachelor of Engineering (B.E.) Capstone Project Dissertation

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Dissertation Executive Summary / Abstract:

Accurate State of Charge (SOC) and State of Health (SOH) estimation is essential for safe, high-efficiency Battery Management Systems (BMS) in electric vehicles and grid storage. Traditional open-loop Coulomb Counting accumulates sensor offset drift, while observer-based Extended Kalman Filters (EKF) impose parameter recalibration bottlenecks and heavy matrix inversion costs on microcontrollers. This dissertation presents an end-to-end cyber-physical alternative using Reservoir Computing via Echo State Networks (ESNs) with fixed random recurrent weights and closed-form linear readout training. The design integrates a 2-RC electro-thermal physics engine, online Recursive Least Squares (RLS) adaptation, C99 firmware with Compressed Sparse Row (CSR) SpMV delivering a 6.7× speedup on ARM Cortex-M4, and a silicon-proven plug-and-play AXI4 IP block targeting Xilinx Artix-7 FPGAs achieving 200/200 bit-exact stage parity with zero DSP slice usage.

Dissertation Chapter Structure:

Chapter 1	**Introduction & Problem Formulation:** Battery management challenges, Coulomb counting drift, and EKF linearization costs.
Chapter 2	**Literature Review:** Physics-based observers (Li et al.), neuromorphic SNNs, and reservoir computing paradigms (Rigutini / Kamarudin).
Chapter 3	**Cyber-Physical Modeling:** 2-RC equivalent circuit, Arrhenius thermal scaling, multi-chemistry engine, and fault injection.
Chapter 4	**Echo State Network Estimator:** Leaky-integrator dynamics, spectral radius control (0.95), and Ridge regression readout training.
Chapter 5	**Embedded Microcontroller Optimization:** C99 firmware, CSR SpMV sparse algebra, 82.4% RAM savings, and Q15 fixed-point LUTs.
Chapter 6	**FPGA Accelerator Architecture:** 100-neuron pipelined RTL, IEEE 1685 IP-XACT AXI4 IP block, and Arty A7-100T bitstream.
Chapter 7	**Experimental Validation:** NASA PCoE battery aging dataset evaluation, 115-test automated regression suite, and bit-exact parity.
Chapter 8	**Conclusion & Industrial Roadmap:** Real-time embedded performance, trade-offs, and commercialization roadmap.